

1. Problem 9 from Section 3.4

$$\vec{x} = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}; \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 1 & -1 & 3 \\ 0 & 2 & 4 \end{array} \right]$$

$$R_2 = R_2 - R_1$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & -2 & 0 \\ 0 & 2 & 4 \end{array} \right]$$

$$R_2 = -R_2/2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 2 & 4 \end{array} \right]$$

$$R_3 = R_3 - 2R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{array} \right]$$

Inconsistent.

2. Problem 11 from Section 3.4

$$\bar{x} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}; \quad \bar{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \bar{v}_2 = \begin{bmatrix} -3 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 2 & 2 & 2 \\ 1 & 3 & 2 \end{bmatrix}$$

$$R_2 = R_2 - 2R_1, \quad R_3 = R_3 - R_1$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 1 & 1/2 \\ 0 & 7 & 5 \end{bmatrix}$$

$$R_1 = R_1 + 3R_2$$

$$\begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 7 & 5 \end{bmatrix}$$

$$R_3 = R_3 - 7R_2$$

$$\begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 5/2 \end{bmatrix}$$

$$v_1 = \frac{1}{2}, \quad v_2 = \frac{1}{2} \Rightarrow [\bar{x}]_B = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

3. Problem 18 from Section 3.4

$$\bar{x} = \begin{bmatrix} 5 \\ 4 \\ 3 \\ 2 \end{bmatrix}, \quad \bar{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \bar{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 1 & 1 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_2 = R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_3 = R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_2 = R_2 + R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_4 = R_4 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -2 \end{array} \right]$$

$$v_1 = 5, v_2 = 3, v_3 = 4$$

$$[\bar{x}]_{B^2} = \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix}$$

Inconsistent, $\bar{x}$ is not in V .

4. Problem 22 from Section 3.4

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \bar{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \bar{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$a. S = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\det S = \cancel{164} -1 -1 = -2$$

$$S^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$B = S^{-1}AS$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1+1 & 1+1 \\ 1-1 & 1-1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 2+2 & 2-2 \\ 0 & 0 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

S. Problem 28 from 3.4

$$A = \begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \quad \bar{v}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \quad \bar{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$\left[\left[\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right]_B \left[\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right]_B \left[\begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right]_B \right]$$

$$\left[\begin{bmatrix} 10-8-2 \\ -8+10-2 \\ -4-4+8 \end{bmatrix}_B \begin{bmatrix} 5+4-0 \\ -4-5-0 \\ -2+2-0 \end{bmatrix}_B \begin{bmatrix} 0-4+4 \\ 0+5+4 \\ 0-2+6 \end{bmatrix}_B \right]$$

$$\left[\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_B \begin{bmatrix} 9 \\ -9 \\ 0 \end{bmatrix}_B \begin{bmatrix} 0 \\ 9 \\ -18 \end{bmatrix}_B \right]$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$